

Designing Accessible XR for Neurodegenerative Disease Patients: Insights from Parkinson's Disease Case Study

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ABSTRACT

XR systems hold transformative potential for healthcare, offering immersive and patient-centered environments that cater to the specific needs of individuals with neurodegenerative diseases. These systems should enhance accessibility, usability, and engagement through intuitive interfaces, multimodal feedback, and controlled environments, improving the quality of data collection while reducing patient anxiety and fostering cooperation during assessments. This paper discusses the accessibility, user experience, and usability of XR system developed using Microsoft HoloLens 2 for Parkinson's Disease patients. It integrates multimodal data collection to evaluate motor functions, speech, cognition, gait, and gaze patterns. Insights from both quantitative metrics and qualitative user feedback provide an understanding of the system's strengths and areas for improvement. The results reveal specific design features that enhance accessibility and usability, such as multimodal guidance and task calibration, while highlighting challenges related to onboarding and motor impairments. These findings emphasize the need for customized XR solutions to address the complex requirements of neurodegenerative patients, showcasing the potential for such systems to support healthcare diagnostics.

Index Terms: Accessibility, Usability, User Experience, Extended Reality, Neurodegenerative Diseases, Parkinson's Disease.

1 INTRODUCTION

Extended reality (XR) technologies have emerged as promising tools for advancing healthcare, particularly in diagnostics and rehabilitation. By creating immersive and interactive environments, XR systems integrate multimodal feedback, including visual, auditory, and tactile cues, to enhance user engagement and focus. These features are especially beneficial for patients with neurodegenerative diseases such as Parkinson's Disease (PD), where traditional diagnostic methods often fail in terms of accessibility and adaptability to the cognitive and motor limitations of patients. Patients with PD face progressive declines in motor control, cognitive function, and speech, which significantly affect their daily lives and quality of life. As users of XR systems, they face unique challenges, including motor impairments, cognitive fatigue, and difficulties with sustained focus. Addressing these challenges requires the design of intuitive interfaces, clear, step-by-step guidance, and task structures that reduce physical and cognitive strain while promoting effective interaction and engagement.

Recent studies highlight the use of XR systems in motor rehabilitation for patients with neurological and neurodegenerative disorders, demonstrating improvements in motor skills, cognitive functions, and patient adherence to therapeutic exercises [2, 10]. Studies on neurodiversity have emphasized the need for intuitive interfaces,

clear instructions, and task designs that minimize physical and cognitive strain to accommodate user limitations [8]. Furthermore, research highlights the importance of creating environments that reduce anxiety and promote participation, particularly for patients who may experience stigma or discomfort in traditional clinical settings [6]. Recent advancements in AR technologies, such as head-mounted displays, have demonstrated potential in addressing these challenges by offering non-intrusive, adaptive interfaces. For example, mixed reality goggles have been used to diagnose Alzheimer's disease [1]. Previously, XR was applied for PD patients for gait analysis [9, 7] or specific interventions like dance intervention [11].

This paper presents an in-depth evaluation of an XR diagnostic system developed for PD patients. Using Microsoft HoloLens 2, the system combines the features of augmented reality with clinically relevant tasks to assess motor functions and cognition [4, 3]. To evaluate the system, we conducted user studies with healthy participants and patients with PD, focusing on accessibility, user experience, and usability. This study provides insights into how XR systems can be optimized for patients with neurodegenerative diseases, which can contribute to a broader understanding of the design of inclusive and effective XR solutions for healthcare applications for elderly patients, with a special focus on PD patients.

2 CASE STUDY

2.1 Design and implementation of the system

The Mixed Reality (MR) diagnostic platform, designed to monitor PD patients [5], uses Microsoft HoloLens 2 to assess motor functions, speech, cognition, gait, and eye movements. It integrates multimodal data acquisition via cameras, microphones, and inertial sensors, operating in a controlled AR environment for consistent, high-precision data collection. Developed with Unity and HoloLens SDK, the system aligns tasks with validated neurological tests, emphasizing accurate data processing and reliable performance. Participants receive feedback through visual, auditory, and textual cues, ensuring clarity and protocol adherence.

2.2 User research

The user research included two groups: 32 healthy controls (HC, 47% women, mean age 59.25 ± 16.52 , 59% higher education) and 15 patients with PD (53% women, mean age 66.14 ± 9.72 , 27% higher education). PD inclusion criteria required a clinical diagnosis of idiopathic Parkinson's disease with mild cognitive impairment or dementia, while exclusion criteria covered severe dementia, vision impairments, and psychotic episodes. All participants provided their informed consent and the protocol was approved by IRB. Sessions lasted 30 min and involved 17 XR tasks that evaluated motor functions, speech, cognition, gait, and gaze patterns using Microsoft HoloLens 2, with step-by-step visual, auditory, and textual guidance. The tasks included activities such as extending arms, touching the nose alternately, repeating specific words or stories, describing images, and following visual prompts with their eyes. Instructions for the tasks were delivered via visual and auditory cues. Quantitative data were collected using the

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Presence Questionnaire (PQ) and System Usability Scale (SUS), analyzed with descriptive statistics and t-tests, while qualitative insights came from interviews and observations.

3 RESULTS AND RECOMMENDATIONS

Enhanced user engagement through multimodal guidance Multimodal guidance, combining visual, auditory, and textual instructions, improved user comprehension, and task execution. Even XR novices navigated easily with little clinician support. Real-time feedback created an engaging, autonomous experience, reducing anxiety. Participants adapted quickly, with task progression boosting motivation. Unlike traditional diagnostic methods, which caused discomfort and self-consciousness due to awareness of voice and motor decline, the system was seen as pleasant, reducing fatigue and maintaining focus. Sequential tasks and real-time feedback enabled smooth progression with minimal external help.

Impact of task design on user fatigue and performance. Task of about 30 min balanced data collection with user comfort. Short, focused tasks maintained attention and minimized fatigue. Despite motor and cognitive impairments, patients completed most tasks successfully. The multi-phase structure provided a sense of accomplishment, sustaining engagement. Initial disorientation in onboarding highlighted the need for clearer introductory instructions.

Usability in healthy and PD participants. Statistically significant differences were observed in SUS scores between HC (80.23 ± 19.37) and PD (61.25 ± 19.37), reflecting usability challenges for PD patients. Significant differences in the PQ sensor fidelity subscale ($p = 0.048$), which measures how well the system's sensory input aligns with users' expectations of realism and responsiveness, highlight the need for tailored XR solutions. Despite lower scores, PD participants appreciated the clarity, structure, and intuitive navigation of the tasks. Challenges like motor difficulties and task orientation diminished as participants progressed.

Cognitive load reduction through simplified interaction methods The system minimized cognitive load with clear instructions, familiar interaction methods (for example, button presses), and intuitive navigation. These features ensured accessibility for all cognitive abilities and reduced frustration. Participants reported comfort and autonomy, finding the experience interactive rather than clinical. Simplified tasks reduced cognitive strain, enabling effective engagement even for technologically inexperienced users.

Onboarding process as a critical determinant of usability Onboarding played a key role in shaping user confidence and usability. Lower SUS scores indicated initial confusion during early interactions, reflected in Q2 (*I found the system unnecessarily complex*, 3.21 ± 1.42) and Q10 (*I needed to learn a lot of things before I could get started with this system*, 2.86 ± 1.46). Despite the admission hurdles, most of the participants expressed readiness to use the system regularly (Q1: *I think that I would like to use this system frequently*, 4.14 ± 0.91). Refinement of tutorials and initial tasks could enhance early user experience and confidence.

Accessibility features enhancing patient comfort and compliance. The system prioritized comfort and accessibility through clear task instructions and minimal physical effort. Directional microphones reduce ambient noise, improving accessibility for users with voice impairments. The AR environment minimized distractions, fostering consistency and reliable data collection.

Perceived autonomy and patient empowerment Participants reported a strong sense of autonomy and control. Independent task navigation and real-time feedback empowered users and reduced the stigma of clinical assessments. PD patients particularly appreciated the engaging, non-clinical environment, which encouraged active participation and reduced reliance on assistance.

Immersion. PQ scores revealed higher immersion for HC participants (5.88 ± 1.62) compared to PD patients (5.28 ± 1.87). Significant differences in sensor fidelity ($p = 0.048$) suggest the need

for improved calibration to ensure consistent performance between user groups. These insights highlight the importance of adaptive sensor configurations in addressing PD-specific limitations.

Longitudinal monitoring and home diagnostics. The system's adaptability supports longitudinal monitoring and home-based diagnostics. Its autonomy reduces reliance on clinicians while ensuring consistent task conditions, making it suitable for remote healthcare, and enabling regular assessments in familiar low-stress settings.

4 CONCLUSION

The case study system investigated in this study demonstrates the feasibility of creating diagnostic tools that blend technological innovation with patient-centric design. Key insights from this study emphasize the critical role of onboarding, task diversity, and real-time feedback in ensuring accessibility and usability across diverse patient populations. Key accessibility features include: 1) clear and step-by-step instructions designed to accommodate possible cognitive impairments in Parkinson's patients, 2) uniform acoustic conditions achieved through Microsoft HoloLens 2 directional microphones, which mitigate ambient noise and improve accessibility for people with voice impairments, and 3) reduced cognitive load through simplified interaction methods, ensuring usability for patients with varying levels of technological familiarity. These findings establish a foundation for further refinement and development of XR applications in healthcare, advancing the creation of more inclusive and effective diagnostic and therapeutic solutions.

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